

1. Project Title

University Management System Network Design using Cisco Packet Tracer

2. Project Overview

This project focuses on the design and implementation of a complete University Management System Network using Cisco Packet Tracer. The network was developed to simulate a real university environment by connecting multiple departments, labs, offices, and server rooms through structured network communication.

The project successfully demonstrates routing, switching, IP addressing, subnetting, and server configuration in a university-wide network.

3. Work Completed

- Designed the complete university network topology
- Configured routers and switches
- Connected all PCs, laptops, printers, and servers
- Implemented IP addressing for all departments
- Established communication between multiple networks
- Configured FTP, DNS, and Web Servers
- Conducted successful connectivity testing
- Verified communication across all departments

4. Final Implementation Status

The project has been fully completed and tested successfully. All devices are connected and configured properly. Communication between departments and servers is functioning correctly.

5. Challenges Faced

- Managing connections between many devices
- Configuring routing paths correctly
- Assigning IP addresses without conflicts
- Troubleshooting communication issues

6. Solutions and Improvements

- Redesigned parts of the topology for better organization
- Corrected routing and interface configurations
- Rechecked IP addressing and subnet assignments
- Performed repeated connectivity testing

7. Technologies and Tools Used

Cisco Packet Tracer, Routers, Switches, FTP Server, DNS Server, Web Server, PCs, Laptops, and Printers

8. Conclusion

The University Management System Network Design project was completed successfully with all required functionalities implemented and tested. The project enhanced practical understanding of networking concepts including routing, switching, subnetting, and server configuration.

9. Reference Topology



